JAN–P–735

18 FEBRUARY 1949

NATIONAL MILITARY ESTABLISHMENT SPECIFICATION

PRIMER, PAINT, ZINC-CHROMATE, ALKYD TYPE

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of the procurement services of the respective departments, and supersedes the following specification:

U. S. Army

Navy Department
52P18
1 June 1945

A. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATIONS

A–1. *Specifications.*—The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

NATIONAL MILITARY ESTABLISHMENT SPECIFICATIONS

JAN–P–106—Packaging and Packing for Overseas Shipment—Boxes, Wood, Nailed.
JAN–B–107—Boxes, Wood, Wirebound (Overseas Type).
JAN–P–124—Packaging and Packing for Overseas Shipment—Containers (Cans, Pails, and Drums), Metal (For Other Than Subsistence Items).

U. S. ARMY SPECIFICATION

100–2—Standard Specification for Marking Shipments by Contractors.[1]

NAVY DEPARTMENT SPECIFICATIONS

General Specifications for Inspection of Material.[2]
52A12—Aluminum-Stearate.
52D7—Driers, Naphthenate, Concentrated, Liquid.
52M2—Magnesium-Silicate (Flatting Extender Pigment).
52R13—Resin, Alkyd, Solution.

FEDERAL SPECIFICATIONS

RR–S–366—Sieves, Standard, Testing.
TT–D–376—Dipentine; for Use in Organic Coatings.
TT–P–141—Paint, Varnish, Lacquer, and Related Materials; General Specification for Inspection, Sampling, and Testing.
TT–S–346—Sienna, Burnt and Raw, Dry (Paint-Pigments).
TT–T–291—Thinner, Paint, Volatile Mineral Spirits (Petroleum-Spirits).
TT–T–425—Titanium-Dioxide, Dry (Paint-Pigment).

[1] Applicable only to Army purchases.
[2] Applicable only to Navy purchases.

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TT–Z–301—Zinc-Oxide, Dry and Paste-in-Oil.
TT–Z–415—Zinc-Yellow (Zinc Chromate); Dry (Paint Pigment).
LLL–B–631—Boxes, Fiber, Corrugated (for Domestic Shipment).
LLL–B–636—Boxes, Fiber, Solid (for Domestic Shipment).

A–2. *Other publication.*—The following publication, of the issue in effect on date of invitation for bids, forms a part of this specification:

BUREAU OF SUPPLIES AND ACCOUNTS PUBLICATION

Navy Shipment Marking Handbook.[3]

B. TYPE

B–1. Paint covered by this specification shall be of but one type as hereinafter specified.

C. MATERIAL AND WORKMANSHIP

C–1. See section E.

D. GENERAL REQUIREMENTS

D–1. See section E.

E. DETAIL REQUIREMENTS

E–1. *Formula.*—

E–1a. Zinc-chromate primer shall consist of ingredients conforming to the requirements of the applicable specifications in the proportions shown in table I.

E–1b. The formula shown in table I is designated Army-Navy-Air Force Paint Formula No. 84/47. Wherever Formula No. 84/47 is specified, the primer shall conform to the requirements of this specification. (See par. H–4.)

TABLE I.—*Formula No. 84/47*

Ingredients	Pounds per 100 gallons
Zinc yellow (spec. TT–Z–415)	270
Titanium dioxide (type II, or III, spec. TT–T–425)	75
Zinc oxide (spec. TT–Z–301, American process)	95
Sienna (type I, spec. TT–S–346)	25
Magnesium silicate (type A or B, spec. 52M2)	75
Aluminum stearate (spec. 52A12)	6
Resin, alkyd, solution (grade I, spec. 52R13) (see par. H–3)	345
Dipentine (spec. TT–D–376)	22
Paint thinner (grade I, spec. TT–T–291)	235
Lead naphthenate drier (type I, spec. 52D7)	12
Cobalt naphthenate drier (type II, spec. 52D7)	1.2
Manganese naphthenate drier (type III, spec. 52D7)	1.2

E–2. *Manufacture.*—The component raw materials shall be mixed and ground as required to produce a product which is uniform, homogeneous, free from grit and entirely suitable for the purpose intended.

[3] Applicable only to Navy purchases.

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Within the limitations imposed by the manufacturing formula, the product shall be capable of being readily broken up with a paddle to a smooth, uniform paint of acceptable consistency, and shall not liver, thicken, curdle, gel nor show any other objectionable properties after long periods of storage.

E–3. *Quantitative requirements.*—Zinc-chromate primer shall conform to requirements shown in table II:

TABLE II.—*Quantitative requirements*

Characteristics	Requirements	
	Minimum	Maximum
Pigment, percent by weight of paint	45.0	49.0
Volatiles, percent by weight of paint	29.5	33.5
Nonvolatile vehicle, percent by weight of paint (calculated by difference)	19.5	23.5
Phthalic anhydride, percent by weight of nonvolatile vehicle	23.0	----
Water, percent by weight of paint	----	.5
Coarse particles and skins (as residue retained on standard 325 sieve (spec. RR–S–366), percent by weight of paint)	----	.5
Viscosity, Krebs units	68	78
Weight per gallon, pounds	11.0	11.6
Time of setting to touch, hours	----	2
Time of drying hard, hours	----	6
Fineness of grind	4	----
Flash point, degrees F	86	----
Zinc oxide (ZnO), percent by weight of pigment	34.0	39.0
Titanium dioxide (TiO_2), percent by weight of pigment	12.0	14.5
Chromium trioxide (CrO_3), percent by weight of pigment	20.0	----

E–4. *Qualitative requirements.*—Zinc-chromate primer shall meet the following qualitative requirements:

E–4a. *Odor.*—The odor shall be normal for the volatiles permitted.

E–4b. *Color.*—The color shall be a clean yellow.

E–4c. *Flexibility.*—The primer shall show no evidence of cracking when subjected to the flexibility test specified in paragraph F–2b (5).

E–4d. *Rosin and rosin derivatives.*—Rosin and rosin derivatives shall not be present.

E–4e. *Compatibility with thinner.*—There shall be no evidence of incompatibility of any of the ingredients of the paint as received when tested as specified in paragraph F–2b (7).

F. METHODS OF SAMPLING, INSPECTION, AND TESTS

F–1. *Sampling.*—

F–1a. *Lot.*—For purposes of sampling primer, a lot shall consist of primer from one batch offered for delivery at one time.

F–1b. *Sampling procedure.*—

F–1b (1). *Ingredient materials.*—When requested by the laboratory officer or other authority, samples of the following ingredient materials shall be taken by the Government inspector for test purposes:

½ pint Zinc yellow.
½ pint Titanium dioxide.
½ pint Zinc oxide.
½ pint Sienna.

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1 pint Magnesium-silicate.
½ pint Aluminum stearate.
1 pint Alkyd resin solution.
1 pint Dipentine.
1 pint Paint thinner.
½ pint Lead naphthenate drier.
½ pint Cobalt naphthenate drier.
½ pint Manganese naphthenate drier.

F–1b (2). *Finished primer.*—Three 1-quart samples shall be taken from each lot by the Government inspector in accordance with the requirements of specification TT–P–141. One sample shall be retained by the contractor, one shall be held by the Government inspector in case of dispute, and one shall be forwarded to the designated testing activity.

F–1c. *Testing activities.*—

F–1c (1). *Army.*—The samples taken by the Government inspector as required for test purposes (see pars. F–1b (1) and F–1b (2)) shall be forwarded to the testing laboratory specified by the contracting officer.

F–1c (2). *Navy.*—The samples taken by the Government inspector as required for test purposes shall be forwarded either to the Laboratory Superintendent, Industrial Test Laboratory, Building No. 121, Philadelphia Naval Shipyard, Naval Base, Philadelphia 12, Pa.; to the Industrial Laboratory, Mare Island Naval Shipyard, Vallejo, Calif., or to the Chemical Laboratory, Norfolk Naval Shipyard, Portsmouth, Va.

F–2. *Tests.*—

F–2a. *Ingredient materials.*—The ingredient materials submitted shall be tested to determine compliance with the requirements of the applicable specifications.

F–2b. *Finished primer.*—

F–2b (1). The following tests shall be conducted in accordance with the applicable methods prescribed in specification TT–P–141:

Tests	Methods
Pigment	402.1[1]
Volatiles	404.1
Nonvolatile vehicle (calculated by difference)	404.1
Water	408
Coarse particles	409
Viscosity	423.1
Weight per gallon	401
Flash point	429.3
Odor	440

[1] Using extraction mixture "A."

F–2b (2). *Phthalic anhydride.*—Phthalic anhydride shall be determined by method 702 of specification TT–P–141. Procedure A shall be employed using a suitable portion of the vehicle, collected during the determination of pigment and evaporated on the steam bath until the volume has been reduced to approximately 10 ml., prior to adding alcoholic KOH. The weight of precipitate obtained shall

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be corected by use of the procedure described by the note appended to procedure. B, method 702.

F-2b (3). *Drying time.*—Drying time shall be determined by method 406.1 of specification TT-P-141, except that the specified conditions of temperature and humidity shall apply only for referee tests in case of dispute. All other tests shall be conducted under prevailing laboratory conditions.

F-2b (4). *Fineness of grind.*—

F-2b (4) a. *Apparatus.*—The device used for measuring fineness of grind shall be as shown on figure 1. This device consists of a steel block and blade. The top surface of the block is ground smooth and contains two parallel ground paths which are ½ inch wide and 2 inches long, being tapered in depth from 0.005 to 0 inch in that length. (The top surface of the block and the bottom of the paths must be surface ground very smooth and true to insure accuracy).

F-2b (4) b. *Procedure.*—To determine the fineness of grind of a paint, place in each path at the deep end a large drop of paint somewhat exceeding the amount required to fill the entire path. Grasp the blade between the thumb and index fingers of both hands. Place the blade edgewise in contact with the surface of the block at the extreme deep end of the paths with the long dimension extending across the width of the block. Hold the blade so that its long dimension is parallel to the end of the block and its short dimension is vertical to the surface of the block. Draw the blade over the surface of the block from the deep end of the path to the shallow end at a uniform moderate rate. Maintain the blade in a vertical position during this operation and continue its motion past the "0" end of the tapered path. (This operation is intended to exactly fill the paths with the paint under test.) *Immediately* inspect the path along the 0 to 8 scale viewing at grazing incidence and note the point where the paint in the path shows a predominantly speckled appearance. Note the number corresponding to the point at which this predominantly speckled appearance begins and report this number as the fineness of grind. Disregard any scattered specks which appear in the path prior to the point where the predominantly speckled appearance is noted. (The reading must be made *immediately* after filling the paths with the paint. For this reason, it is often advisable to make a preliminary determination to establish the approximate position of the first appearance of predominantly speckled surface. A second accurate reading can be made very rapidly since the approximate position is already known).

F-2b (5). *Flexibility.*—

F-2b (5) a. *Panel.*—The panel used shall be cut from bright tin plate weighing not more than 25 gm. nor less than 19 gm. per square decimeter (0.51 to 0.39 pound per square foot). Commercial No. 31 gage bright tin plate should weigh about 0.44 pound per square foot. The panel shall be 5 inches long and not less than 3 inches wide, and shall be free of sharp bends or dents. Edges of the panel shall be lightly sanded to remove burrs due to cutting.

F-2b (5) b. *Film applicator.*—The applicator used shall be of such construction and dimensions that the thickness of the resulting dried film of the paint under test is 0.0010 ± 0.0003 inch. (An applicator which has been successfully used is shown on figure 2 of this specification.) The applicator shall be made of suitable nonmagnetic material.

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F-2b (5) c. *Magnetic chuck.*—The magnetic chuck shall be of the permanent magnet type. The area of the working surface on the chuck shall be greater than the area of the panel and the chuck shall have sufficient magnetic strength to hold the tin panel in contact with the chuck throughout the area of the panel.

F-2b (5) d. *Procedure.*—The handle of the magnetic chuck shall be adjusted to the "off" position. A sheet of paper (newspaper is suitable) shall be laid on the chuck. The tin panel shall be placed on the paper and handle adjusted to the "on" position. The panel shall now be cleaned with benzol or other suitable solvent. A small quantity of paint shall be placed on one end of the panel and a film spread by means of the applicator described in paragraph F-2b (5) b. The handle shall be adjusted to the "off" position, the panel removed and allowed to air dry 2 hours in a horizontal position and then baked 24 hours in an air circulating oven at 100° to 105° C. At the end of the baking period the panel shall be allowed to stand for 30 minutes at 25 ± 5° C. and then bent at its midpoint through 180 degrees over a ⅛-inch mandrel. The painted surface of the panel shall be uppermost during the bending which shall be accomplished at a uniform rate over a period of approximately two seconds. The panel shall be examined at the bend using a 5-power lens and any cracking shall be noted.

F-2b (6). *Rosin and rosin derivatives.*—Test for rosin and rosin derivatives shall be conducted according to method 503.1 of specification TT-P-141. A portion of the separated nonvolatile vehicle shall be used for the test.

F-2b (7). *Compatibility.*—Compatibility with thinner shall be determined in accordance with method 489 of specification TT-P-141. Fifty ml. of primer and 50 ml. of paint thinner conforming to the requirements for grade I of specification TT-T-291 shall be used. Observations shall be made immediately after mixing and also 30 minutes after mixing.

F-2b (8). *Pigment analyses.*—

F-2b (8) a. *Preliminary separation.*—Weigh out 2.000 gm. sample of the pigment and transfer to a 250 ml. beaker. Moisten the pigment with acetone and add 25 ml. of 1 : 4 sulfuric acid. Let stand at room temperature for an hour with occasional stirring. Filter, wash with 1 : 4 sulfuric acid at room temperature until free of chromium, and transfer filtrate to a 250 ml. volumetric flask. Reserve residue for determination of titanium dioxide. Dilute filtrate to exactly 250 ml. and reserve for zinc and chromium determinations.

F-2b (8) b. *Zinc oxide.*—

F-2b (8) b (1). Remove 50.0 ml. portion from the volumetric flask and transfer to a 400 ml. beaker. Dilute to 150 ml. with distilled water and add 25 ml. of a 25 percent solution of tartaric acid. Make just neutral to litmus paper with ammonium hydroxide and add 25 ml. of formic acid mixture. (See par. F-2b (8) b (2).) (Care should be observed to adjust the acidity to the proper pH for quantitative precipitation of the zinc as sulfide.) Pass a rapid stream of hydrogen sulfide through the solution for 30 minutes. Filter off the precipitated zinc sulfide and wash with distilled water saturated with hydrogen sulfide. Place the precipitate and paper into a tared porcelain crucible, dry the paper, and then char at low heat until the paper is

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consumed. Increase the heat to 1,000° C. and ignite for 10 minutes. Let cool in desiccator and weigh as zinc oxide.

F–2b (8) b (2). *Reagent formic acid mixture.*—To 400 ml. formic acid add 60 ml. ammonium hydroxide. Dissolve 500 gm. of ammonium sulfate in approximately 1 liter of distilled water. Mix the two solutions and add sufficient water to make 2 liters.

F–2b (8) c. *Chromium trioxide.*—Remove a 50.0 ml. portion from the volumetric flask. Transfer to a 600 ml. beaker. Dilute to 300 ml. with distilled water and add 15 ml. of concentrated sulfuric acid and 3 ml. of concentrated nitric acid. Bring to a boil, then add 1 ml. of 2.5 percent silver nitrate solution and 1 ml. of 0.1N potassium permanganate. Slowly add 10 ml. to 20 percent ammonium persulfate, freshly prepared. The pink color of permanganate should persist after 10 minutes of boiling, adding more ammonium persulfate if necessary and boiling 10 minutes after last addition of ammonium persulfate. Five ml. of 1:3 hydrochloric acid shall be added and the solution boiled for 10 minutes after the permanganate color is destroyed. Cool to 20° C. and add standardized ferrous ammonium sulfate solution (approximately 0.1N) until approximately 5 ml. in excess has been added. Now titrate with approximately 0.1N potassium permanganate solution (which has been standardized against sodium oxalate) until an excess of approximately 5 ml. has been added. Adjust to the correct end point by careful addition of the standardized ferrous ammonium sulfate solution. Multiply the volume of permanganate solution used by the ferrous ammonium sulfate equivalent of 1 ml. of standardized permanganate solution, subtract the product from the amount of ferrous sulfate used and calculate the amount of chromium as CrO_3. To determine the ferrous ammonium sulfate equivalent, take as much of the ferrous ammonium sulfate as was used in the test, dissolve or dilute in a solution having the same volume and acidity, titrate as above with the permanganate solution, and calculate from the data obtained.

F–2b (8) d. *Titanium dioxide.*—Place filter paper and residue (see paragraph F–2b (8) a) in a 300 ml. Erlenmeyer flask. Add 20 ml. of concentrated sulfuric acid, cover the flask with a watch crystal and heat on a hot plate until the paper is well charred. Add concentrated nitric acid drop by drop until the solution is light yellow in color. Remove from hot plate, cool, and add 3 ml. of concentrated nitric acid, 1 ml. of perchloric acid (60 percent) and 10 gm. ammonium sulfate. Place on hot plate and heat until strong fumes of sulfur trioxide are evolved. Place the flask on a gauze supported by a tripod and boil gently over a flame until all soluble material has been dissolved (about 20 minutes should be sufficient). Cool and dilute with 200 ml. of water. Pass a rapid stream of hydrogen sulfide through the contents of the flask until the antimony has been completely precipitated. Filter, and wash the precipitate and paper thoroughly with cold 5 percent sulfuric acid which has been saturated with hydrogen sulfide. Boil the filtrate until all hydrogen sulfide has been removed. Cool. Add 10 ml. of ammonium hydroxide (sp. gr. 0.90) to lower the acidity. Add 0.1 normal potassium permanganate solution until a slight permanent pink color is produced. This usually does not require more than one or two drops of 0.1 normal potassium permanganate solution. Transfer to a 250 ml. volumetric flask and adjust to volume. Remove 100 ml. and reduce as follows:

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F–2b (8) d (1). *Procedure.*—Pour 20 ml. of zinc amalgam into a 250 ml. separatory funnel fitted with a rubber stopper in place of a ground stopper. Add solution to be reduced to the funnel and then displace air with CO_2 by passing gas into funnel for 5–10 seconds. Replace stopper and shake for 5 minutes. (Some work has shown that 2–3 minutes shaking will be sufficient. The use of a shaking machine with an automatic timer is recommended for this operation.) Remove rubber stopper and wash carefully. Wash sidewalls of funnel and add 5 ml. CCl_4. Draw off the amalgam and a minimum quantity of CCl_4, since this tends to foul the amalgam. (The amalgam is drawn off into the storage funnel for revivification and any necessary cleaning.) Add 1 ml. saturated NH_4CNS and titrate with standard ferric ammonium sulfate solution to an end point which persists for 30 seconds.

F–2b (8) d (2). *Preparation of liquid zinc amalgam.*—Pour 1,600 gm. Hg into a porcelain casserole and add 40 gm. of zinc. Add 30 ml. 1:4 H_2SO_4 and heat on hot plate at 70° to 80° C. for ½ hour or until all zinc has dissolved in the mercury. Cool and transfer to a separatory funnel. Add two short pieces of stick zinc and 5–10 ml. 10 percent H_2SO_4. Stopper funnel and keep until needed. (This quantity of amalgam is adequate for 4–8 reductors.) The amalgam becomes fouled with use and this is shown by dirty appearance. The amalgam shall be washed by shaking in the separatory funnel with 10 ml. 10 percent H_2SO_4 for 2 minutes. The amalgam shall be separated from the dilute acid and is ready for use. With proper care in separation of CCl_4 from amalgam it is necessary to wash the amalgam after 40–50 determinations, but with poor separation of CCl_4 it is necessary after 5–10 determinations. The amalgam should be stored over a beaker containing water to catch any droplets of mercury remaining in the stem of the funnel. The use of a funnel rack is recommended for storing of shaking funnels and storage funnel. The stock liquid amalgam should stand for 5 minutes after shaking to allow any granules of zinc to rise to the surface, otherwise these granules give trouble during drawing off of amalgam from solution by blocking stopcock.

F–2b (8) d (3). *Ferric ammonium sulfate (0.005 gm. TiO_2/ml.*—For 18 liters of solution, weight 544 gm. of c.p. ferric ammonium sulfate ($24H_2O$) into a 4 liter breaker. Add 3,500 ml. distilled water and then 200 ml. conc. H_2SO_4. Stir until in solution. Carefully add (dropwise) 0.1N $KMnO_4$ until a very faint pink color is obtained. Filter and dilute to 18 liters. Mix well. Bureau of Standards TiO_2 No. 154 is used for standardizing the ferric ammonium sulfate solution.

F–3. *Acceptance of lots.*—Lots shall be accepted by the Government inspector only upon receipt of a satisfactory test report from the laboratory on the sample primer and when necessary, on samples of ingredients.

F–4. *Additional inspection.*—Where other specifications form a part of this specification, unless otherwise specified, sampling, inspection, and tests shall be conducted as required in the referenced specifications.

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT

G–1. Packaging and packing requirements for domestic shipment apply only for direct Government purchases. Unless otherwise

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specified, packaging and packing requirements for domestic shipment to subcontractors, contractors, or shipbuilders may be in accordance with best commercial practice.

G–2. *Packaging (for domestic and overseas shipment).*—Primer shall be furnished in 1-gallon multiple-friction-top cans or in 5-gallon lug-cover steel pails, as specified. (See par. H–2.)

G–2a. One-gallon multiple-friction-top cans shall conform to the requirements for type IV of specification JAN–P–124.

G–2b. Five-gallon lug-cover steel pails shall conform to the requirements for type VI of specification JAN–P–124.

G–3. *Packing.*—

G–3a. *For domestic shipment.*—

G–3a (1). Four 1-gallon cans shall be packed in a nailed wood or wirebound box conforming to the requirements of specifications JAN–P–106 or JAN–B–107, respectively, or in a style RSC corrugated or solid fiberboard box having a minimum Mullen test of 275 pounds and conforming to the requirements of specifications LLL–B–631 or LLL–B–636, respectively. No strapping will be required.

G–3b (2). Five-gallon lug-cover steel pails will require no overpacking.

G–3b. *For overseas shipment.*—

G–3b (1). Four 1-gallon cans shall be packed in a nailed wood or wirebound box conforming to the requirements of specification JAN–P–106 or JAN–B–107, respectively.

G–3b (2). Five-gallon lug-cover steel pails will require no overpacking.

G–4. *Marking.*—

G–4a. *One-gallon cans.*—One-gallon cans shall be durably and plainly marked with the following information completed:

Specification No. _____
Material _____
Stock No. (as applicable) _____
Quantity _____
Contractor _____
Manufacturer _____
Contract or Order No. _____
Date of Manufacture _____

G–4b. *Steel containers and shipping containers.*—Steel containers and shipping containers shall be plainly marked with the following information completed:

Material _____
Quantity _____
Contractor _____
Specification No. _____
Contract No. _____
Gross Weight _____

In addition to the foregoing, shipments for the Army shall be marked in accordance with the requirements of U. S. Army Specification 100–2; for the Navy, in accordance with the requirements of the Navy Shipment Marking Handbook.

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H. NOTES

H–1. *Use.*—Zinc chromate primer covered by this specification is intended for use as a corrosion-inhibiting primer on metal surfaces.

H–2. *Ordering information.*—Requests, requisitions, schedules, and contracts or orders should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Size of container required. (See par. G–2.)
(*c*) Whether the subject commodity is to be prepared for domestic or overseas shipment. (See sec. G.)

H–3. If it is desired to use alkyd resin solution conforming to the requirements for grade II of specification 52R13, the weight of the alkyd resin solution, as specified in this specification, should be multiplied by 1.17, and the paint thinner, petroleum spirits (grade 1), reduced by 0.17 times the weight of alkyd resin solution.

H–4. For information only, the formula of table I is based on a theoretical 103 gallons (approx.) to allow for normal manufacturing loss. Composition by volume is approximately as follows:

Ingredients:	*Gallons per 100 gallons*
Zinc yellow	9.2
Titanium dioxide	2.3
Zinc oxide	2.0
Sienna	0.9
Magnesium silicate	3.2
Aluminum stearate	0.7
Alkyd resin	43.8(29.0)[1]
Dipentine	3.1
Paint thinner	36.0
Lead naphthenate drier	1.2
Cobalt naphthenate drier	0.15
Manganese naphthenate drier	0.15
Total	102.7

[1] Figure in parentheses refers to volume of resin solid (nonvolatile).

H–5. Zinc chromate primer should be purchased by volume, the unit being a U. S. gallon at 15.5° C. (60° F.).

H–6. Copies of National Military Establishment specifications and Federal specifications (required for Army purchases) and U. S. Army specifications may be obtained, as indicated in the "Index of the United States Army, Joint Army-Navy and Federal Specifications and Standards." Copies of this index may be obtained from the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Services within the Department of the Army will obtain copies of National Military Establishment, U. S. Army, and Federal specifications through established departmental channels. Both the title and identifying number or symbol should be stipulated when requesting copies.

H–7. Copies of National Military Establishment specifications and Federal specifications (required for Navy purchases), Navy Department specifications, and the Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval

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Supply Center, Norfolk 11, Va. Both the title and identifying number or symbol should be stipulated when requesting copies.

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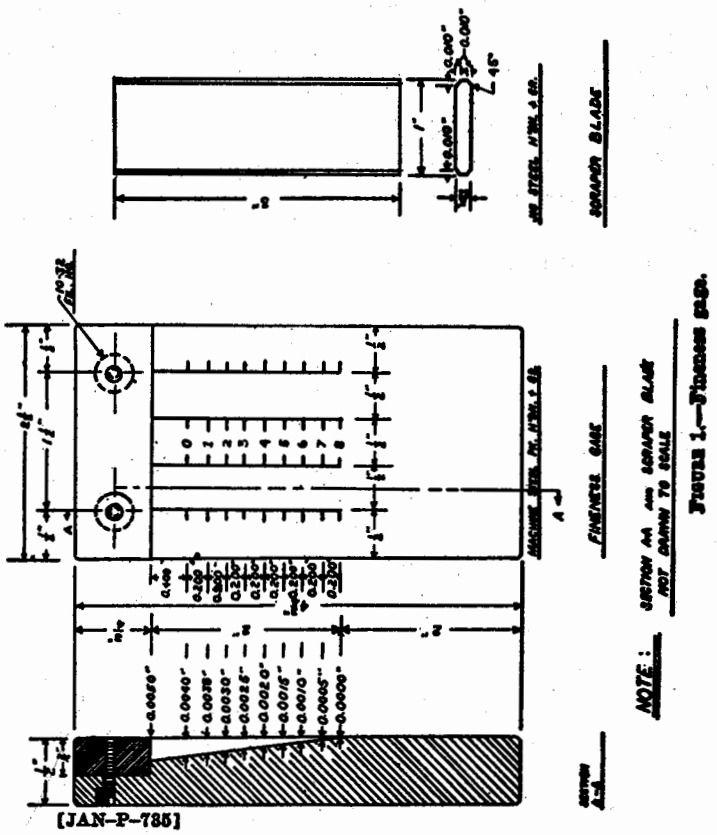


FIGURE 1.—Fineness gage.

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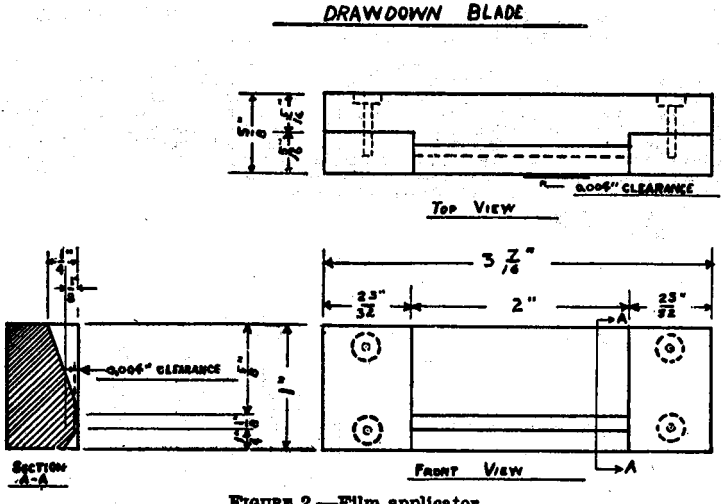


FIGURE 2.—Film applicator.

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